

Task: Build a Real-Time Client Project Dashboard with Role-Based Access & Live Activity Feed

Overview full stack web application that a small agency uses internally to manage client projects, track task progress, and monitor team activity in real time. This is not a tutorial CRUD app — it requires WebSocket implementation, role-based permission logic, and state management decisions that require genuine engineering judgment.

What You Must Build

1. Authentication & Role System

Three roles with strictly different access levels:

Role	Access
Admin	Full access — manage clients, projects, users, view all activity
Project Manager	Create & manage projects, assign tasks, view their team's activity only
Developer	View assigned tasks only, update task status, cannot see other developers' tasks

- JWT-based authentication (access token + refresh token)
- Refresh token must be stored in an HttpOnly cookie — not localStorage
- Role middleware must be enforced on every protected route at the API level — frontend-only role hiding is not acceptable
- A Developer must not be able to reach a Project Manager's data even by directly hitting the API endpoint with a modified token

2. Project & Task Management

- Admin and PM can create projects and assign them to clients

- Projects contain tasks — each task has: title, description, assigned developer, status (To Do / In Progress / In Review / Done), priority (Low / Medium / High / Critical), due date, and an activity log
- Task status changes must be recorded with a timestamp and the user who made the change — this log must be stored in the database, not derived
- PM can only manage projects they created — they cannot see or edit another PM's projects
- Tasks past their due date must be automatically flagged as Overdue — this must happen via a scheduled background job, not on page load

3. Real-Time Activity Feed

This is the core technical challenge of the task.

- Implement a live activity feed using WebSockets (Socket.io or native WebSocket — your choice, justify it in the README)
- When any user updates a task status, all users currently viewing that project must see the update in real time without refreshing
- The feed must show: who made the change, what they changed, and when — formatted as "Ravi moved Task #12 from In Progress → In Review · 2 mins ago"
- Admin sees activity across all projects in a single global feed
- PM sees activity only from their own projects
- Developer sees activity only on tasks assigned to them
- If a user is offline and comes back, they must see the last 20 activity events they missed — this must be fetched from the database, not cached in memory

4. Dashboard & Filters

- Admin dashboard: total projects, total tasks by status, overdue task count, active users online right now (shown as a live count using WebSocket presence)
- PM dashboard: their projects summary, tasks by priority, upcoming due dates this week
- Developer dashboard: their assigned tasks, sorted by priority then due date
- All task lists must support filtering by status, priority, and due date range — filters must work via query parameters so they are shareable as URLs

5. Notifications

- When a task is assigned to a developer, they receive an in-app notification (stored in DB, shown in UI)
- When a task they own is moved to In Review, the PM receives a notification
- Notifications must show as a count badge and expand into a dropdown — mark as read individually or all at once
- Unread notification count must update in real time via WebSocket, not polling

Technical Requirements

- **Frontend:** React with TypeScript — no plain JavaScript accepted
- **Backend:** Node.js with Express or Fastify — your choice, justify it
- **Database:** PostgreSQL — use proper relational schema with foreign keys, indexes on frequently queried columns, and explain your indexing decisions in the README
- **ORM:** Prisma or raw SQL — no MongoDB, no NoSQL
- **Real-time:** WebSocket — no long-polling, no SSE
- **Background Jobs:** Use node-cron or Bull queue for the overdue task scheduler — justify your choice
- **Validation:** All API inputs must be validated server-side — frontend validation alone is not sufficient
- **Error Handling:** All API endpoints must return consistent, structured error responses — no raw stack traces exposed to the client
- **Environment:** All secrets in .env — never hardcoded

Seed Data Required Your repository must include a seed script that creates:

- 1 Admin, 2 Project Managers, 4 Developers
- At least 3 projects with 5+ tasks each in various statuses
- At least 2 tasks already in overdue state
- Pre-existing activity log entries so the feed is not empty on first load

Submission Requirements

- Public GitHub/GitLab repository
- Host the application on Vercel and share the live application link
- README must include: local setup instructions (Docker preferred), database schema diagram or description, architectural decisions (WebSocket library choice, job queue choice, token storage approach), known limitations
- In the Explanation field (150–250 words): the hardest problem you solved, how you handled the real-time role-filtered feed, and one thing you'd do differently

Evaluation

Criteria	Weight
Role-based access — enforced at API level, not just frontend	25%
Real-time feed — correct, role-filtered, with missed event catchup	25%

Database design — schema quality, relationships, indexing	20%
Code architecture — separation of concerns, TypeScript usage	20%
Seed data, README, setup experience	10%

Auto-Disqualification: Role access enforced only on frontend, WebSocket replaced with polling, no seed raw script, no TypeScript, SQL mixed randomly into controllers, hardcoded secrets, missing refresh token implementation.

Completed the task? Submit your work here: (<https://bit.ly/4bGXmZV>)